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Exp No :2 DEVELOPING AND DEPLOYING A SIMPLE CHAINCODE

Aim:

To develop, package, install, approve, commit, and execute a simple asset transfer chaincode in a **Hyperledger Fabric test network** using the **JavaScript programming language**.

Requirements:

- Linux Operating System (Kali)
- Docker and Docker Compose
- Go Programming Language
- Git and Curl
- Hyperledger Fabric Samples, Binaries and Docker Images

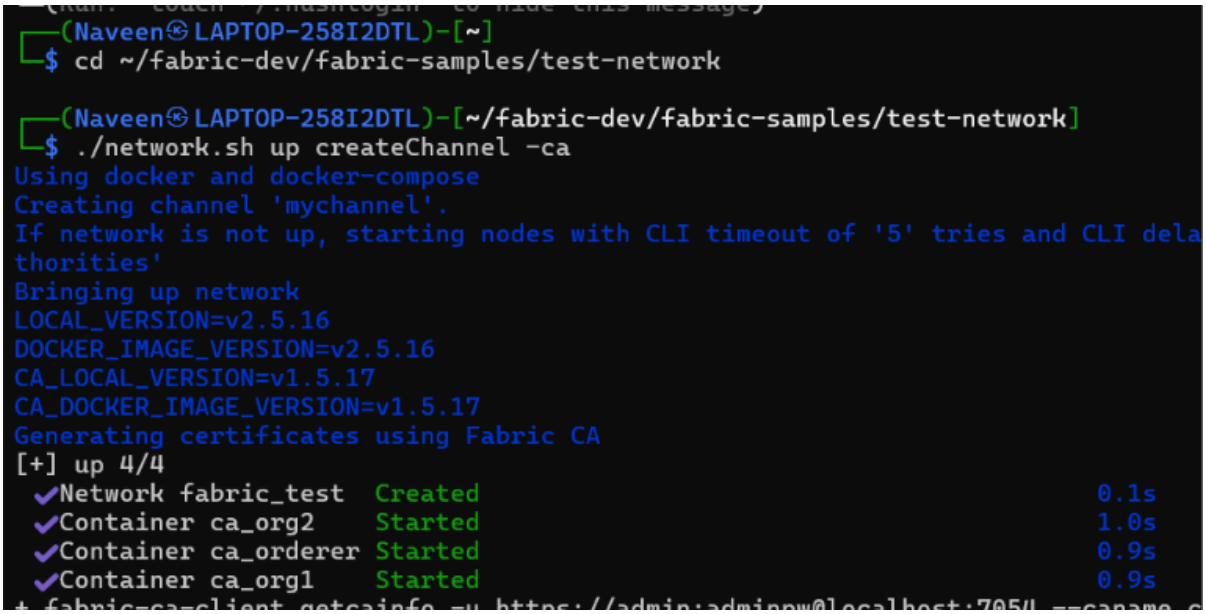
Procedure:

1. Navigate to the Test Network

cd ~/fabric-dev/fabric-samples/test-network

2.Start the Fabric Network

./network.sh up createChannel -ca



3. Deploy the JavaScript Chaincode

./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript

```
┌──(Naveen㉿LAPTOP-258I2DTL)-[~/fabric-dev/fabric-samples/test-network]
└─$ ./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-javascript --
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
+ peer lifecycle chaincode queryinstalled --output json
+ jq -r 'try (.installed_chaincodes[].package_id)'
+ grep '^basic_1.0:1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629$'
+ test 1 -ne 0
+ peer lifecycle chaincode install basic.tar.gz
+ res=0
2026-09-01 20:23:01.309 IST 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal -> Installed rem
445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629\022\tbasic_1.0" >
2026-09-01 20:23:01.310 IST 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal -> Chaincode cod
c307bc2b5fbe664c87b24c035bee15cbcc7f59b629
Chaincode is installed on peer0.org1
```

4. Verify Installed Chaincode

peer lifecycle chaincode queryinstalled

```
┌──(Naveen㉿LAPTOP-258I2DTL)-[~/fabric-dev/fabric-samples/test-network]
└─$ peer lifecycle chaincode queryinstalled
Installed chaincodes on peer:
Package ID: basic_1.0:1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629, Label: basic_1.0
```

5. Initialize the Ledger

peer chaincode invoke -o localhost:7050 \

--ordererTLSHostnameOverride orderer.example.com \

--tls --cafile "$ORDERER_CA" \

-C mychannel -n basic \

-c '{"function":"InitLedger","Args":[]}'

```
┌──(Naveen㉿LAPTOP-258I2DTL)-[~/fabric-dev/fabric-samples/test-network]
└─$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile "$ORDERER_CA" -C mychannel -n basic -c '{"functi
on":"InitLedger","Args":[]}'
2026-09-01 20:34:20.249 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
```

6. Query All Assets

peer chaincode query -C mychannel -n basic \

-c '{"Args":["GetAllAssets"]}'

```
(Naveen㉿LAPTOP-258I2DTL)-[~/fabric-dev/fabric-samples/test-network]
$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

```
(Naveen㉿LAPTOP-258I2DTL)-[~/fabric-dev/fabric-samples/asset-transfer-basic/chaincode-javascript]
$ sed -n '16,35p' lib/assetTransfer.js
async InitLedger(ctx) {
    const assets = [
        {
            ID: 'asset1',
            Color: 'blue',
            Size: 5,
            Owner: 'Tomoko',
            AppraisedValue: 300,
        },
        {
            ID: 'asset2',
            Color: 'red',
            Size: 5,
            Owner: 'Brad',
            AppraisedValue: 400,
        },
        {
            ID: 'asset3',
            Color: 'green',
            Size: 10,
```

7. Create a New Asset

peer chaincode invoke -o localhost:7050 \

--ordererTLSHostnameOverride orderer.example.com \

--tls --cafile "$ORDERER_CA" \

-C mychannel -n basic \

-c '{"function":"CreateAsset","Args":["asset10","Black","15","Arun","800"]}'

8. Query the Newly Created Asset

peer chaincode query -C mychannel -n basic \

-c '{"Args":["ReadAsset","asset10"]}'

9. Transfer Asset Ownership

peer chaincode invoke -o localhost:7050 \

--ordererTLSHostnameOverride orderer.example.com \

--tls --cafile "$ORDERER_CA" \

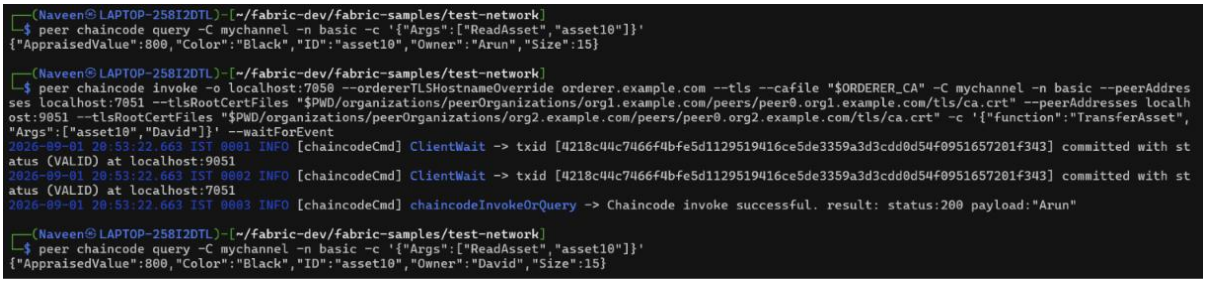
-C mychannel -n basic \

-c '{"function":"TransferAsset","Args":["asset10","David"]}'

10. Verify Updated Asset

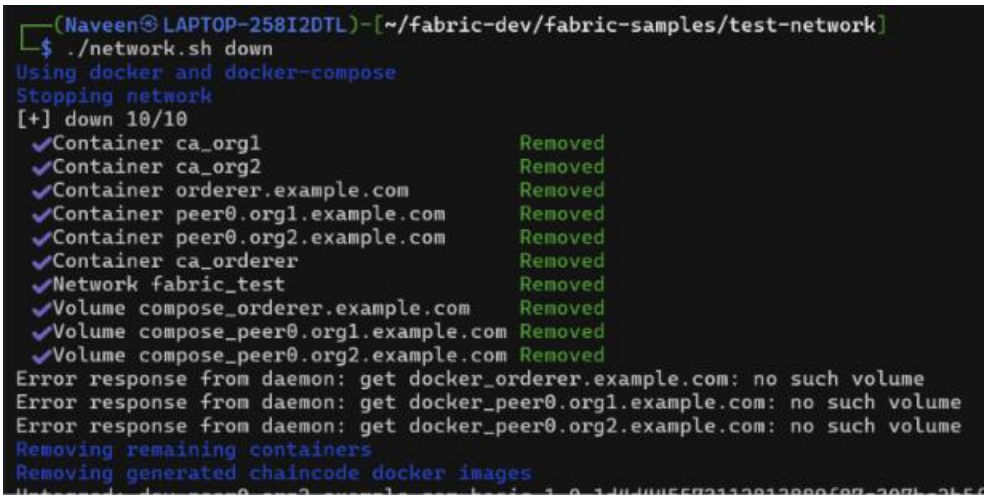
peer chaincode query -C mychannel -n basic \

-c '{"Args":["ReadAsset","asset10"]}'



11. Stop the Network

./network.sh down



Result

The simple **JavaScript chaincode was successfully developed, deployed, installed, approved, committed, and executed** on the Hyperledger Fabric test network. Asset records were successfully created and queried, and ownership of **asset10** was successfully transferred from **Arun to David**, demonstrating the basic lifecycle of a smart contract in Hyperledger Fabric.